HEIQ XMEMBRANE PROJECT

Prior material search consolidation

What was already learned · what is a real lead · what must be measured next

Internal team report · 31 August 2026

This report consolidates the complete local material-search record under `/home/torsheim/HeiQ`. It does not restart the search. It separates named membranes and literature leads from counter-ion routes, synthetic inverse-design coordinates, historical screening constraints and current same-sample qualification requirements.

Correct phase statement. Considerable material work has already been completed. It established a Li^+-form cation-exchange/ionomer route, identified several named evidence anchors, tested broad hypothetical property spaces and derived architecture-dependent targets. What remains is to convert that work into traceable supplier and physical-coupon records, then measure the coupled property bundle on the same specimens. This is a consolidation and qualification phase, not a new search from zero.

Local scope indexed	39,284 files and 1,039 PDF files across five HeiQ roots plus the top-level utility
Downloads record retained	45 manifest rows, 31 unique PDFs and 14 duplicate copies
Traceable source set	75 canonical current, historical, archive and workspace records with SHA-256 hashes
Named material/reference records	10; none has a complete same-sample MAT-01–MAT-09 bundle
First evidence-anchor coupon	Selemion CMVN in Li^+ form, if construction and supply are confirmed
Controls	Nafion NR211/NR212 only as method and thickness controls, not target materials

Abstract

The local record contains extensive previous material work, but it is spread across literature registers, ion screens, Nafion references, inverse-design studies, downloaded reports and thousands of synthetic model coordinates. The work did not identify a commercially validated "optimal membrane." Its most defensible real-material result is Selemion CMVN: in liquid single-salt tests at 295 ± 1 K, reported water drag was 1.1 ± 0.8, 9.2 ± 0.8, 4.9 ± 0.2 and 3.7 ± 0.4 mol H_2O per mol charge for H^+, Li^+, Na^+ and K^+ forms. This makes Li^+ the best same-platform counter-ion hypothesis and Selemion the strongest local mechanism coupon, not a demonstrated system-performance candidate. CR61 AZL 386, MF-4SC, Nafion 117/120 and BPSH contribute useful property fragments; microporous polyamine, plaster–silica, perforated ion-exchange and graded or adsorbing composites remain literature/architecture leads.

The historical V12–V15 sequence supports the Li^+ route but also shows why it cannot be promoted directly to procurement ranking. V12's cross-study prior gave Li^+ a median conditional flux of 0.442 kg,m^{-2},h^{-1} and SEC of 0.125 kWh/kg at 0.6 V. V13 identified wet vapour conductivity as the dominant information gap, followed by drag and diffusivity. V14's supported 25-μm checkpoint remained conditional, and V15 reduced its bulk performance once two vapour interfaces were introduced; favourable-gradient flux was about 2.68 times the opposing-gradient result. The current direct-EO and active-ERV studies likewise define sampled property regions, not material recipes.

The earlier transport numbers above 100 are fully retained: they originated in hypothetical V4 and later inverse-design sweeps extending to $t_w = 150$, while the current Terminal-1 vector has $t_w = 100.0608$. Neither is a measured material. No named local candidate has a complete, same-sample set of sorption, chemical water mobility, EO drag, wet through-plane conductivity, compressed-stack ASR, swelling/thickness, two-sided flux, interface compatibility and provenance. The immediate step is therefore an availability-and-coupon shortlist built from the existing leads, followed by one controlled protocol on the same physical samples. Candidate ranking and net energy claims remain open until that evidence exists.

The governing V8 decision sequence makes post-ERV O30 the first system-relevant coupon target. It removes 1,228.27 kg/h on the declared boundary and has sampled solutions at 1.00–1.25 V for every one-factor $D = 10^{-14}$–10^{-9} m^2/s coordinate, although the conditional power window deteriorates from $+162.416$ to -181.413 kW. Direct EO inside the ERV is a separate secondary architecture: 178/320 rows close transport, 50/320 pass the engineering screen and 0/320 pass the full coupled gate. Neither screen identifies a physical membrane or a net system saving.

Decision. Do not perform another broad hypothetical material sweep. Begin supplier/availability consolidation and coupon qualification using the existing search results. Start with the Li^+-form CEM/ionomer route and a Selemion CMVN evidence anchor, retain non-PFSA and microporous/composite routes, and use Nafion only as a reference control.

1 Audit scope and evidence classes

EVERYTHING LOCAL IS RETAINED, BUT NOT EVERYTHING IS EQUIVALENT EVIDENCE

The consolidation inventories the five active local roots: `HeiQ-Thermodynamical-Analysis`, `HeiQ_V2`, `XMembrane_Model`, `XMembrane_Trond_Study` and `deliverables`. It also includes the hashed manifest of PDFs previously reviewed in Windows Downloads. Frozen report copies are retained for chronology and reproducibility, while top-level source, configuration and output files remain the maintained implementation chain. Git metadata, backup/cache directories and this newly generated handoff directory are excluded from the counts to avoid self-inflation. The single top-level file, `collect_for_chatgpt.sh`, was also inspected; it is a historical three-project bundle utility, not material evidence. Byte-identical project and top-level V8/V8.1 release mirrors are present in the physical file count but are represented once in the evidence register.

Current authority. V8.1 is the latest hash-frozen controlled English delivery; all 107 declared checksums and the recorded 300/300 live tests pass. It is an additive translation layer with no scientific delta from V8, not an immutable whole-repository archival snapshot. The governing science and decisions remain the V8 Current Result and Decision Report; the model-closure report is supporting provenance where V8 does not differ. This consolidation is an internal post-release addendum and does not mutate or enter the V8/V8.1 allowlist.

The rebuilt weekly Revision 1 now includes the post-release direct-ERV weather figure and uses the same authority hierarchy. It remains a dated, non-governing presentation; it adds no named material or full-system closure.

Evidence tier	Meaning	Current examples
E0	Concept or unqualified architecture hypothesis	Graded, MOF-buffered or adsorbing composite; symmetric-AC rectification route
E1	Named literature family or mechanism	BPSH, microporous polyamine, plaster–silica, perforated ion-exchange architecture
E2	Named grade/product with unmatched property fragments	Selemion CMVN, CR61, MF-4SC, Nafion 117/120 and current Nafion controls
E3	Partial bundle measured on one identifiable lot/coupon	None in the local evidence
E4	Complete same-sample MAT-01–MAT-09 coupon bundle	None
E5	Repeated batch and subsystem evidence	None

The phrase “validated Terminal-1 vector” in older material means numerically reproduced. It does not mean experimentally measured. Likewise, a model row named “candidate,” “optimum” or `Mxxxxx` is not a physical material unless identity, grade, batch and source are independently present.

2 What the previous material search established

THE SEARCH PRODUCED USEFUL ROUTE KNOWLEDGE AND TARGET HISTORY

Workstream	Retained finding	Correct present use
V4	Hypothetical water-transport-number sweep to $t_w = 150$	Explains historical $t_w > 100$; no named material demonstrated it
Legacy material-sweep V8/V11	2,304 synthetic property combinations and 12 retained M-ID regression vectors	Parameter-space and code-regression history only; unrelated to controlled Release V8
V9/V10	Nafion-relative inverse envelopes and decomposition of effective transport	Search vocabulary; properties were deliberately independent
V12	Literature-bounded H/Li/Na/K comparison ranked Li^+ most favourably	Select counter-ion route, not a product
V13	Thin Li^+ optimum at the minimum-thickness boundary; conductivity dominated value of information	Prioritize vapour conductivity, then drag and D/sorption measurements
V14	Supported 25-μm Li^+ checkpoint with added ASR	Coupon architecture checkpoint, not a mechanical recommendation
V15	Two-interface vapour model reduced performance and exposed gradient-direction sensitivity	Measure both films and both gradient directions
V42–V55	Successive inverse, wet-receiver and surface-robustness screens	Preserve chronology; V53 is the relevant $\leq$ 2-V descendant, while V55 required 3.177 V
V61	Direct-EO-in-ERV inverse frontiers	Historical integrated-ERV requirements; area-conditioned and synthetic
V62	Practical break-even $t_w \approx 18.43/27.39/35.85/44.82$ for O60/O50/O40/O30	Architecture-dependent target envelope with receiver SEC excluded
Current post-ERV	O30 common-voltage screen below 2 V	First system-relevant coupon replay; conditional budget, not net saving
Current direct ERV	Terminal-1 gradient, $+J/0/-J$ and coupled-gate replays	Secondary/open architecture; replay only after passive transfer and films are measured

The evolution matters. Early studies explored deliberately independent properties to learn which combinations might work. Later studies introduced hydration correlations, receiver surfaces, two gas films, voltage budgets and fuller energy ledgers. A point that passed an intermediate screen could fail a later closure: the V43 M00653/70-times-PFSA-drag point, for example, disappeared under the V44 robust-surface test. The correct response is to retain the point as provenance, not carry it forward as a material lead.

3 Named materials and literature leads

REAL IDENTITY FIRST; PERFORMANCE CLAIMS SECOND

Material or family	Local quantitative evidence	Use now	Class
Selemion CMVN	Drag H/Li/Na/K = $1.1/9.2/4.9/3.7$ with reported uncertainties in liquid at 295 K	First mechanism/evidence-anchor coupon; compare ion forms from one lot and recover vapour bundle	P0 test priority, not ranking
CR61 AZL 386	Approximate aqueous drag H $\sim$ 2, K $\sim$ 11	Secondary counter-ion reference if construction and supply are practical	Secondary test lead
MF-4SC	Fully hydrated conductivity H/Li/Na = $2.8/0.61/0.62$ S/m	Alkali-PFSA conductivity lead; grade and matched properties missing	Family evidence lead
Nafion 117	Li/Na conductivity $1.3/1.1$ S/m at 98% RH; nominal thickness 178–183 μm	Thick counter-ion literature bridge only	Control
Nafion 120	Liquid-equilibrated $D_w = 2.65 \times 10^{-10}$ (Na) and 2.15×10^{-10} m^2/s (K)	Diffusivity reference fragment; do not rank on D alone	P2 reference
BPSH	Vapour H^+-drag source, no complete local vector	Non-PFSA sulfonated aromatic search family	P2 family lead
NR211/NR212	25/50 μm PFSA references; matched NR211 active row is 91.63% diffusive and uses 0.493 kW more electricity than passive	Method, thickness and negative/mechanism-control coupons only	Controls
Microporous polyamines	Very high apparent water-per-charge behaviour in liquid/anionic literature	Explore reciprocal definitions and vapour applicability; not an intrinsic t_w input	P1 exploratory
Plaster–silica gel	Direct EO-dehumidification precedent in Gerlach 2008	Recover construction and operating conditions	P1 literature recovery
Perforated/porous IEM	Yaroshchuk asymmetric EO architecture	Separate architecture and possible rectification route	P2 architecture
Graded/adsorbing/MOF composite	Local conceptual proposal only	Search taxonomy after a base ionomer is characterized	P2 exploratory

No structured local commercial matrix was found for Aquivion, Fumasep, Flemion, Aciplex, Gore, SPEEK, PBI, PVA, chitosan or cellulose membranes. This does not prove that those families are unsuitable. It means the existing local work did not produce traceable candidate rows for them, so they belong in the optional targeted gap-filling list after availability checks on the existing named leads; they must not trigger another broad desk search or be silently treated as screened out.

Test and evidence-anchor order. (1) Li^+-form CEM/ionomer and a supported thin-layer variant; (2) Na^+-form CEM/ionomer; (3) non-PFSA sulfonated aromatic and microporous chemistry; (4) K^+ as a mechanism control; (5) porous/perforated or graded composite architectures. H-form PFSA grades are controls, not the desired optimum.

4 Quantitative route evidence and present target gaps

ARCHITECTURE CHANGES THE REQUIRED MEMBRANE REGION

4.1 The Li^+ route from V12 to V15

Study	Median flux [kg,m^{-2},h^{-1}]	Median SEC [kWh/kg]	Interpretation
V12 Li^+, 100 μm, 0.6 V	0.442	0.125	Best cross-study ion prior; not a real joint distribution
V13 inverse Li^+, 15 μm, 0.6375 V	3.57	0.131	Minimum-thickness boundary; conductivity drives uncertainty
V14 supported Li^+, 25 μm, 0.675 V	2.35	0.136	Adds 0.05 mΩ,m^2 support/interface ASR
V15 opposing gradient 65→80% RH, 0.55 V	0.434	0.217	Two vapour interfaces erase much of the bulk-only advantage
V15 favourable gradient 80→60% RH, 0.60 V	1.164	0.142	About 2.68 times the opposing-gradient flux

These numbers compare different conditional models and must not be interpreted as one development trajectory with identical system boundaries. Their robust shared result is the experimental priority: retain the Li^+ hypothesis, but measure vapour-state conductivity, EO drag, sorption/mobility and both interfaces on the same specimen.

4.2 Named evidence versus current model coordinates

Comparison	Evidence value	Model coordinate	Reading
Selemion Li drag vs direct EO	9.2 ± 0.8 in liquid	Preferred sampled cores around $t_w = 20$ or 25–30	Central value is 2.17 times below $t_w = 20$; conditions do not match
Selemion Li drag vs active ERV	9.2	Terminal-1 $t_w = 100.0608$	10.88-fold gap before D, σ, sorption and films
Upper historical Li prior vs active ERV	15	$t_w = 100.0608$	6.67-fold gap; still a cross-study prior
MF-4SC Li conductivity vs active ERV	0.61 S/m, fully hydrated	Terminal-1 $\sigma_{0.8} = 23.1719$ S/m	38-fold warning-scale gap under non-matched conditions
Nafion drag envelope vs active ERV	0.3–2.5	$t_w = 100.0608$	At least 40-fold gap; confirms reference-only role

The active-ERV Terminal-1 coordinate is

$$t_w = 100.0608, \quad D_{\mathrm{ref}} = 1.598 \times 10^{-14}\ \mathrm{m^2/s}, \quad \sigma_{\mathrm{ref}} = 23.1719\ \mathrm{S/m}, \quad L = 36.9137\ \mu\mathrm{m}. \tag{1}$$

It is a synthetic inverse target, not "the optimal membrane." In addition, the physically relevant passive term is not scalar D by itself but the chemical-permeance bundle

$$K_{a,w} \propto \frac{D(a_w, T)}{L} \frac{\partial C_w}{\partial a_w}, \tag{2}$$

together with drag, conductivity/ASR and source/receiver films. A favourable ERV gradient can assist passive transport before the pinch, so blindly minimising permeance may also remove useful transfer.

> Do not build a fictitious candidate by taking drag from Selemion, conductivity from MF-4SC, diffusivity from Nafion 120 and thickness from an inverse-design row. That combination has never existed on one membrane. The previous studies used such combinations deliberately as hypothesis space; the next phase must test whether the required properties coexist.

4.3 Governing experiment sequence and the two distinct O30 boundaries

Model boundary	Current result	Material implication
Post-ERV O30	1,228.27 kg/h; sampled common-bus solutions at 1.00–1.25 V for all seven D coordinates	First system-relevant 25-cm^2 coupon target; measure a joint property curve, not D alone
Post-ERV D sensitivity	Known power rises 30.055 to 373.885 kW and conditional window falls $+162.416$ to -181.413 kW from 10^{-14} to 10^{-9} m^2/s	High chemical permeance can erase the conditional energy room; one-factor result only
Post-ERV full remaining duty	0/49 sampled points confirmed; 21 physical no-solutions and 28 numerical/local unresolved	No demonstrated $\leq$ 2-V solution, but not a global no-go proof
Direct-ERV O30	4,365.918 kg/h at the selected diagnostic; bulk direction favourable, surface margin -0.00797	Needs 1.137-times film capacity/diagnostic area for zero surface margin and 1.733-times for $+0.02$; never merge with post-ERV O30
Direct-ERV full duty	Zero-bulk-gradient pinch; no finite film/area multiplier creates positive surface margin at finite flux	Surface direction remains a diagnostic rather than an active-EO feasibility gate
Direct-ERV symmetric $\pm J$	Incremental even rate 0.001518 kg/h at O30 (0.348 ppm of $+J$) and zero at full duty	Present quasi-static constant-drag model gives no useful AC rectification; frequency and asymmetry remain untested hypotheses
Direct-active coupled gates	178/320 transport-closed, 50/320 engineering passes, 0/320 full-coupled passes	Secondary/open architecture, not the first procurement or coupon target
Post-release direct-ERV weather replay	Fixed inverse area 34,425.05 m^2; O80/O50/O30 voltages 0.2500/0.1491/0.04155 V and totals 107.856/93.293/88.235 kW	O80/O50 pass the declared engineering screen, O30 fails stack/engineering and all full-system claims are false; not in V8.1 or material evidence

The passive ERV+AC reference is 543.741 kW. The post-ERV O30 subtotal of 381.327 kW leaves a conditional 162.414-kW model budget, not 162.414 kW of demonstrated saving. Receiver/sink energy, EO heat rejection, electrodes, manifolds, leakage and durability remain outside the closed ledger. The direct-active diagnostic is still more conditional and cannot replace this first test sequence. The post-release weather replay uses three different outdoor duties and a weather-specific inverse voltage root, not a manufactured fixed-stack off-design model or a same-duty savings comparison.

5 Qualification plan: turn old search results into candidates

ONE PHYSICAL ID, ONE BATCH, ONE CONDITIONING HISTORY

The candidate master should contain physical products or synthesized batches only. Terminal-1, M-IDs, V43–V62 coordinates and direct-grid exemplar IDs belong in a separate inverse-design scenario table. Every physical candidate must retain supplier, product grade, lot/sample, ion form, conditioning, temperature, humidity definition, thickness and source provenance.

Gate	Measurement	Why it blocks ranking
MAT-01	Sorption, desorption and hysteresis	Defines loading, $\partial C_w/\partial a_w$ and chemical work
MAT-02	Water D/mobility and chemical permeance	Quantifies passive assistance or back-diffusion with an explicit definition
MAT-03	EO water drag	Defines polarity-odd water per charge under the actual ion form and RH
MAT-04	Wet through-plane conductivity	Sets membrane ohmic work in the same hydration state
MAT-05	Membrane and compressed-stack ASR	Exposes contacts/support/interface penalties missing from bulk conductivity
MAT-06	Dry/wet thickness, swelling and mechanics	Couples permeability, resistance, durability and supported-layer design
MAT-07	Coupled two-sided net water flux and charge	Tests whether all favourable properties coexist and closes the water balance
MAT-08	Receiver/electrode/interface compatibility	Determines reversibility, leakage, corrosion and receiver burden
MAT-09	Provenance, grade and availability	Makes the result procurable, repeatable and rankable

5.1 Recommended first coupon matrix

1. Confirm whether Selemion CMVN of known construction and lot is obtainable. Prepare Li^+, Na^+, K^+ and H^+ forms from the same lot where chemically defensible. Record sheet dimensions, support/open area, minimum order and the number of usable 80-by-80-mm coupons. This is the cleanest local counter-ion test, not a performance ranking.
2. Run NR211 and NR212 in the identical fixture as thin/thickness PFSA controls. They validate the method; they do not define the target.
3. Add one obtainable non-PFSA sulfonated aromatic/BPSH-like membrane and one microporous/adsorbing architecture coupon when identity and construction are traceable.
4. Use split common inlet, foil blank, zero-current positive/negative gradients and ABBA/BAAB polarity blocks. Close source loss against receiver gain; repeat masks, gap and flow; keep Δp within ± 5 Pa and run separate 0, ± 20 and ± 50 Pa pressure challenges.
5. Use the same physical coupon for sorption, mobility, drag, conductivity and stack ASR wherever the method permits. Freeze replicate count only after the pilot and conditioning-yield check.
6. Replay measured property curves first at post-ERV O30 and only then in direct active-ERV if measured passive transfer and films justify it. Report solved current density and area, but apply neither a 21-A/m^2 limit nor an exploratory area acceptance cap.

The canonical starting fixture has a 50-by-50-mm maximum active aperture, approximately 80-by-80-mm coupon, 110-by-110-mm outer plates and interchangeable 1/2/4-mm symmetric gaps, with 2 mm nominal. Six L/min per side is the film-characterisation mode, not a universal measurement flow: the declared low-signal detectability limits are 0.69 and 2.29 L/min per side, so use a separately calibrated low-flow mode or integrating water balance. The active channel contains 5.00 mL and proposed non-membrane dead volume is 1.222 mL per side; both fixture dimensions and loads are starting values to be measured. A quiescent or shallow closed cell can support transient inventory work, but it cannot represent a fixed-boundary steady state without an explicit sink and boundary evolution. The 25-cm^2 aperture is a laboratory fixture choice, not a system-area cap.

6 Conflicts and superseded rules retained explicitly

NO SILENT HARMONISATION

Conflict	What differs	Required treatment
Two Nafion implementations	Older local model uses 1,980 kg/m^3, polynomial D and $n_d = \mathrm{clip}(2.5\lambda/22)$; newer library uses 1,970 kg/m^3, Motupally D and drag scenarios 0.3/1/2.5	Name the implementation and conditioning; never merge into a same-sample vector
N117 thickness	Historical 178 μm versus current grade library 183 μm	Report 178–183 μm with source/condition
Terminal-1 aliases	`terminal1_anchor`, `terminal1_post_erv_anchor`, `legacy_030_case_94_material` and `Terminal1_optimum` are the same synthetic vector	Map to one scenario record; not a material
`030_final_material`	Comparator label lacks a matched local property-source row	Do not auto-alias to Terminal-1
M14424/M14425	Legacy material-sweep V8 report and atlas disagree on an extreme-row ID	Preserve both until the raw grid row resolves the label
OFAT versus global conductivity	Local high-conductivity saturation appears weak while broad screens rank conductivity strongly	State sampled range and reference point; this is scale dependence, not a contradiction
Standalone D	Older studies rank scalar/PFSA-relative D; current closure uses $D(\partial C_w/\partial a_w)/L$	Keep old D work as one-factor sensitivity only
MAT identifier collision	Historical MAT-01–MAT-05 and current MAT-01–MAT-09 assign different properties to the same IDs	Join by property meaning and schema/source version, never MAT ID alone
Relocated report manifests	Report-level relative paths no longer resolve after copying into V8/V8.1, although top release hashes pass	Use top MANIFEST for copy integrity and original canonical manifests for scientific provenance
Frozen report copies	Code/data copied into report bundles duplicate maintained top-level sources	Use for report reproduction, not as independent implementations or evidence

The following constraints are also superseded as present acceptance rules: 20- or 100-m^2 area gates; a 21-A/m^2 current-density gate; the historical 2,000-A/m^2 dense-ionomer screen as a product limit; 5-V material sweeps; Nafion as the target; positive passive surface direction as an active-EO feasibility gate; and low D as a sufficient material criterion. Preferred material work remains below 2 V, while area and current density are reported as engineering diagnostics.

7 Decision and concrete next move

THE SEARCH IS CONSOLIDATED; CANDIDATE RANKING IS NOT YET EARNED

Decision item	Status	Action
Previous material-search knowledge recovered	COMPLETE	Use the generated named-material, literature-lead, route, target, alias and source registers
Latest controlled delivery checked	COMPLETE	V8.1 verifies 107/107 hashes and declares no scientific delta from governing V8
Start from zero	NO	Preserve V4–V62 and current model results as the search basis
Lead chemistry/ion direction	READY	Li^+-form CEM/ionomer first; retain Na^+ and non-PFSA/microporous alternatives
First evidence-anchor coupon	READY	Check Selemion CMVN supply/construction and prepare ion-form controls if obtainable
First system target	READY	Post-ERV O30 in the 25-cm^2 controlled-flow fixture
Direct EO inside ERV	OPEN	Secondary replay only after measured passive transfer and film behaviour; present full-coupled gate is 0/320
Commercial procurement ranking	OPEN	Populate supplier, grade, lot, thickness, availability and E2 evidence; no current row qualifies as a lead solution
Measured material ranking	OPEN	Complete MAT-01–MAT-09 on identifiable same-sample groups
Net system energy claim	NO-GO	Requires measured receiver, heat, fan/manifold, electrode, leakage and durability ledger
Additional broad hypothetical sweeps	STOP	Only replay measured candidates or a named subsystem uncertainty

Immediate team task. Build an availability sheet from the existing leads, not a fresh unstructured material search. For each candidate, record supplier, exact grade, ion form/exchange route, construction/support, thickness, sheet dimensions, usable 80-by-80-mm coupon count, lot/sample availability and source. Then select a small coupon set that maximizes mechanism discrimination: Selemion CMVN ion forms, NR211/NR212 controls, one non-PFSA sulfonated candidate and one microporous/adsorbing or porous-architecture lead. No candidate should be called “optimal” until the same-sample bundle is measured and replayed.

8 Machine-readable handoff and canonical evidence

THE REPORT IS A VIEW; THE REGISTERS ARE THE WORKING INTERFACE

The handoff was generated by `scripts/build_material_search_handoff_v1.py`. The local evidence contract passes, including source existence, source-ID resolution, the 2,304 row legacy material-sweep V8 grid, the V4 $t_w = 150$ maximum, the four V62 thresholds, the two Nafion implementations, both controlled-release checksum sets and the 45-row Downloads manifest. Eight targeted tests pass.

- `data/named_material_register.csv`: physical/named material and control records;
- `data/literature_lead_register.csv`: chemistry and architecture leads not yet promoted to products;
- `data/candidate_availability_sheet.csv`: pre-populated availability, identity and first-question worklist;
- `data/development_route_register.csv`: Li/Na/K, direct-EO and direct-active-ERV routes;
- `data/synthetic_target_register.csv`: Terminal-1, M-IDs and other non-material model coordinates;
- `data/qualification_matrix.csv`: 90 candidate-by-MAT evidence cells;
- `data/gap_to_current_targets.csv`: quantitative gap comparisons with their condition caveats;
- `data/historical_screen_milestones.csv`: retained V4–V62 and current search chronology;
- `data/model_conflict_register.csv` and `data/alias_register.csv`: no-silent-merge rules;
- `data/source_inventory.csv`: 75 source paths, sizes and SHA-256 hashes;
- `data/handoff_summary.json` and `data/output_manifest.csv`: decision and generated-file audit.

The governing current energy, gradient and decision interpretation remains in the V8 Current Result and Decision Report delivered unchanged through V8.1. The model-closure report is supporting provenance. This material addendum neither changes that claim boundary nor promotes conditional post-ERV or active-ERV power differences to net savings.

> Permitted wording is: *the previous studies identify a prioritized material and ion-family search space, with Selemion CMVN as the strongest local evidence-anchor coupon and Li^+ as the leading counter-ion hypothesis.* Do not state that an optimal membrane, a complete procurement shortlist or a net energy-saving material has been identified.

End of prior-material-search consolidation. All conclusions refer to the local evidence state indexed on 31 August 2026.